

1. What is Artificial Intelligence?

Artificial Intelligence (AI) is a branch of computer science that enables machines and computers to perform tasks that normally require human intelligence. AI systems can learn from data, recognize patterns, solve problems, make decisions, and understand language. Nowadays, AI is used in many areas such as education, healthcare, banking, transportation, and entertainment. Examples of AI applications include chatbots, voice assistants, recommendation systems, and self-driving technologies. AI helps improve efficiency and makes many daily tasks easier and faster.

2. Difference between AI, Machine Learning, Generative AI, and Agentic AI

Artificial Intelligence (AI) is the broad concept of creating machines that can perform intelligent tasks similar to humans. Machine Learning (ML) is a subset of AI that allows computers to learn from data and improve their performance without being explicitly programmed for every task. Generative AI is another branch of AI that focuses on creating new content such as text, images, videos, music, and code. Tools like ChatGPT and image generators are examples of Generative AI. Agentic AI is a more advanced form of AI that can independently plan actions, make decisions, and perform tasks to achieve a specific goal with minimal human involvement. In simple terms, AI is the overall field, Machine Learning helps systems learn from data, Generative AI creates new content, and Agentic AI can act and make decisions on its own.

3. Explain any three open-source AI models available on Hugging Face

GPT-2

GPT-2 is an open-source language model developed by OpenAI. It is capable of generating human-like text based on the input provided by users. It can be used for text generation, content writing, summarization, and answering questions.

Stable Diffusion

Stable Diffusion is a popular open-source image generation model. It creates images from text descriptions entered by users. This model is widely used for creating artwork, illustrations, and creative designs using AI.

DeepSeek-R1

DeepSeek-R1 is an open-source AI model designed for reasoning, coding, and problem-solving tasks. It can assist users in programming, mathematics, and logical reasoning. The model is becoming popular because of its strong performance and open-source availability.

4. What is the difference between traditional programming and AI-assisted development?

Traditional programming is a software development approach where programmers write all instructions manually. The computer follows the exact rules and logic written by the developer. If the instructions are incorrect, the program may not work as expected. In traditional programming, every condition and rule must be clearly defined by the programmer.

AI-assisted development is a modern approach where AI tools help developers write code, suggest solutions, find errors, and improve productivity. Instead of writing every line manually, developers can use AI to generate code suggestions and automate repetitive tasks. AI-assisted development helps save time and can make software development faster and easier. However, developers still need to review and verify the generated code for accuracy.

5. Explain AI hallucinations with one example

AI hallucination is a situation where an AI system generates information that appears correct but is actually false or inaccurate. This happens because AI models predict responses based on patterns in data rather than truly understanding facts. As a result, they may sometimes create incorrect information with confidence.

For example, if a user asks an AI tool for details about a book, the AI might provide a wrong publication date or even mention an author who never wrote the book. The answer may look convincing, but it is not factually correct. Therefore, users should always verify important information generated by AI systems. AI hallucinations are one of the major challenges in the development and use of AI technologies.

